



elastic

Observability metrics

Observability metrics give insights into how an application is behaving. They help us understand performance, health, and behavior of systems by looking at the data they produce. observability lets you look at telemetry data and figure it out.

In modern software systems, observability usually means collecting and analyzing telemetry data like:

- metrics
- logs
- traces
- profiles

4 pillars of observability

Traditionally observability is explained using three pillars.

1. Metrics

Metrics are numerical info collected over a set period of time They show system behavior and performance trends like

- CPU usage
- memory usage
- response time

Metrics are useful because they make it easy to monitor systems and create alerts.

2. Logs

Logs are records of events and incidents that occur in a system. They are timestamped and give detailed information about what happened.

Different types of logs exist:

- System logs
- Application logs
- Network logs

Logs help us understand why errors or incidents happen

3. Traces

Traces show how a request travels through a system.

In modern distributed systems, a single request goes through many services.

Tracing tracks the whole path for us to investigate

This helps us detect where the problem is happening

4. Profiles

It analyzes stack traces and resource usage to find performance issues in code.

The 4 golden signals (metrics) for SRE teams

Latency

Latency measures the time it takes for data to travel from one point to another. Latency will signal underlying performance issues. High latency can degrade user experiences by increasing load times

Traffic

Traffic metrics track the volume of requests an application processes. They help SRE teams understand user behavior and anticipate scaling needs.

Errors

Error metrics provide visibility into failed requests or operations. Monitoring error rates and identifying patterns can help address recurring issues.

Saturation

Saturation metrics indicate how close a system is to its capacity limits.